

# Social Media Platform Case Study Assignment (Replica Set)

## Collections and Columns

### users

| Column         | Type   | Description     |
|----------------|--------|-----------------|
| userId         | String | Unique user ID  |
| username       | String | Username        |
| email          | String | User email      |
| bio            | String | Profile bio     |
| followersCount | Number | Total followers |
| followingCount | Number | Total following |

### posts

| Column     | Type   | Description  |
|------------|--------|--------------|
| postId     | String | Post ID      |
| userId     | String | Linked user  |
| caption    | String | Post caption |
| hashtags   | Array  | Tags used    |
| likesCount | Number | Total likes  |
| createdAt  | Date   | Post time    |

### comments

| Column      | Type   | Description       |
|-------------|--------|-------------------|
| commentId   | String | Comment ID        |
| postId      | String | Linked post       |
| userId      | String | Comment user      |
| commentText | String | Comment           |
| commentedAt | Date   | Comment timestamp |

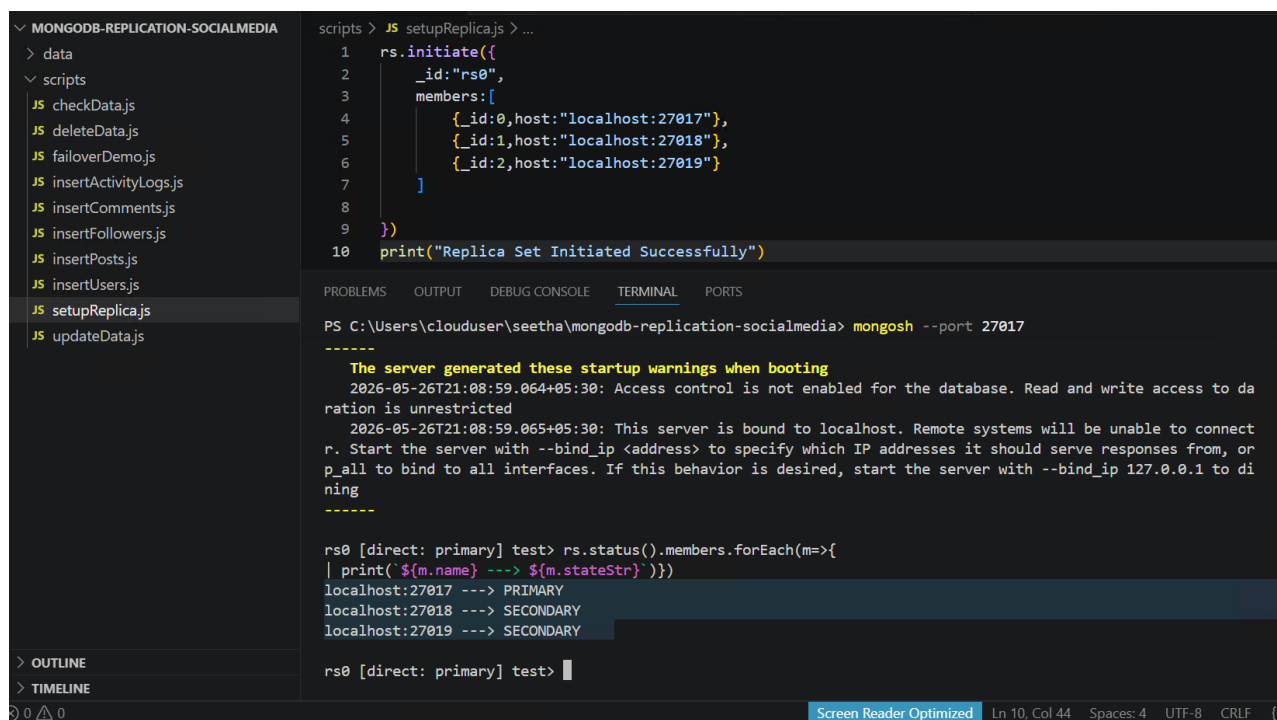
### followers

| Column      | Type   | Description      |
|-------------|--------|------------------|
| followId    | String | Follow ID        |
| followerId  | String | Follower user    |
| followingId | String | Following user   |
| followedAt  | Date   | Follow timestamp |

activity\_logs

| Column       | Type   | Description        |
|--------------|--------|--------------------|
| activityId   | String | Activity ID        |
| userId       | String | User reference     |
| activityType | String | Login/Post/Like    |
| device       | String | Mobile/Laptop      |
| activityTime | Date   | Activity timestamp |

## REPLICA SET CONFIGURATION



The screenshot shows a code editor with a file explorer on the left and a terminal on the right. The file explorer shows a project named 'MONGODB-REPLICATION-SOCIALMEDIA' with a 'scripts' folder containing several JavaScript files. The 'scripts' folder is expanded, showing files like 'checkData.js', 'deleteData.js', 'failoverDemo.js', 'insertActivityLogs.js', 'insertComments.js', 'insertFollowers.js', 'insertPosts.js', 'insertUsers.js', 'setupReplica.js', and 'updateData.js'. The 'setupReplica.js' file is selected, and its content is visible in the editor. The terminal shows the output of running 'mongosh --port 27017', displaying startup warnings and the status of the replica set members.

```
scripts > JS setupReplica.js > ...
1  rs.initiate({
2    _id:"rs0",
3    members:[
4      {_id:0,host:"localhost:27017"},
5      {_id:1,host:"localhost:27018"},
6      {_id:2,host:"localhost:27019"}
7    ]
8  })
9
10 print("Replica Set Initiated Successfully")
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017
-----
The server generated these startup warnings when booting
2026-05-26T21:08:59.064+05:30: Access control is not enabled for the database. Read and write access to da
ration is unrestricted
2026-05-26T21:08:59.065+05:30: This server is bound to localhost. Remote systems will be unable to connect
r. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or
p_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to di
ning
-----

rs0 [direct: primary] test> rs.status().members.forEach(m=>{
| print( `${m.name} ---> ${m.stateStr}` )})
localhost:27017 ---> PRIMARY
localhost:27018 ---> SECONDARY
localhost:27019 ---> SECONDARY

rs0 [direct: primary] test> |
```

0 0 0 Screen Reader Optimized Ln 10, Col 44 Spaces: 4 UTF-8 CRLF

# CREATE

MONGODB-REPLICATION-SOCIALMEDIA

> data

> scripts

JS checkData.js

JS deleteData.js

JS failoverDemo.js

JS insertActivityLogs.js

JS insertComments.js

JS insertFollowers.js

JS insertPosts.js

JS insertUsers.js

JS setupReplica.js

JS updateData.js

scripts > JS insertUsers.js > ...

```
1 use("socialmedia")
2 db.users.insertMany([
3   { userId:"U1", username:"john_doe", email:"john@gmail.com", bio:"Photography lover", followersCount:500,
4     { userId:"U2", username:"jane_smith", email:"jane@gmail.com", bio:"Travel enthusiast", followersCount:100,
5     { userId:"U3", username:"ravi_kumar", email:"ravi@gmail.com", bio:"Foodie and chef", followersCount:200,
6     { userId:"U4", username:"priya_raj", email:"priya@gmail.com", bio:"Fitness freak", followersCount:1200,
7     { userId:"U5", username:"alex_93", email:"alex@gmail.com", bio:"Music producer", followersCount:300, f
8   ]})
9 print("Users inserted successfully!")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017

rs0 [direct: primary] test>

(To exit, press Ctrl+C again or Ctrl+D or type .exit)

rs0 [direct: primary] test>

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertUsers.

Users inserted successfully!

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertFollow

Followers inserted successfully!

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertCommen

Comments inserted successfully!

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertPosts.

Posts inserted successfully!

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertActivi

Activity logs inserted successfully!

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>

○ PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>

> OUTLINE

> TIMELINE

# READ

## (From checkData.js)

MONGODB-REPLICATION-SOCIALMEDIA

> data

> scripts

JS checkData.js

JS deleteData.js

JS failoverDemo.js

JS insertActivityLogs.js

JS insertComments.js

JS insertFollowers.js

JS insertPosts.js

JS insertUsers.js

JS setupReplica.js

JS updateData.js

scripts > JS checkData.js

```
1 use("socialmedia")
2 print("=== Users ===")
3 printjson(db.users.find().toArray())
4 print("\n=== Posts ===")
5 printjson(db.posts.find().toArray())
6 print("\n=== Comments ===")
7 printjson(db.comments.find().toArray())
8 print("\n=== Followers ===")
9 printjson(db.followers.find().toArray())
10 print("\n=== Activity Logs ===")
11 printjson(db.activity_logs.find().toArray())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\checkData.js

=== Posts ===

[

{

\_id: ObjectId('6a15c01949a19d550fab114'),

postId: 'P1',

userId: 'U1',

caption: 'Beautiful sunset!',

hashtags: [

#sunset,

#photography

],

likesCount: 150,

createdAt: ISODate('2024-01-10T00:00:00.000Z')

},

{

\_id: ObjectId('6a15c01949a19d550fab115'),

postId: 'P2',

> OUTLINE

> TIMELINE

## (From port 27017-primary)

▼ MONGODB-REPLICATION-SOCIALMEDIA

> data

▼ scripts

JS checkData.js

JS deleteData.js

JS failoverDemo.js

JS insertActivityLogs.js

JS insertComments.js

JS insertFollowers.js

JS insertPosts.js

JS insertUsers.js

JS setupReplica.js

JS updateData.js

> OUTLINE

> TIMELINE

scripts > JS insertUsers.js > ...

1 use("socialmedia")

2 db.users.insertMany([

3 { userId:"U1", username:"john\_doe", email:"john@gmail.com", bio:"Photography lover", followersCount:500, followingCount:300 },

4 { userId:"U2", username:"jane\_smith", email:"jane@gmail.com", bio:"Travel enthusiast", followersCount:200, followingCount:100 },

5 { userId:"U3", username:"ravi\_kumar", email:"ravi@gmail.com", bio:"Foodie and chef", followersCount:150, followingCount:80 },

6 { userId:"U4", username:"priya\_raj", email:"priya@gmail.com", bio:"Fitness freak", followersCount:120, followingCount:60 },

7 { userId:"U5", username:"alex\_93", email:"alex@gmail.com", bio:"Music producer", followersCount:180, followingCount:90 },

8 ])

9 print("Users inserted successfully!")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017

rs0 [direct: primary] test> use socialmedia

switched to db socialmedia

rs0 [direct: primary] socialmedia> show collections

activity\_logs

comments

followers

posts

users

rs0 [direct: primary] socialmedia> db.users.find()

[

{

\_id: ObjectId('6a15bf815ae0cd9242abc114'),

userId: 'U1',

username: 'john\_doe',

email: 'john@gmail.com',

bio: 'Photography lover',

followersCount: 500,

followingCount: 300,

}

]

## (From port 27018-secondary)

▼ MONGODB-REPLICATIO...

> data

▼ scripts

JS checkData.js

JS deleteData.js

JS failoverDemo.js

JS insertActivityLogs.js

JS insertComments.js

JS insertFollowers.js

JS insertPosts.js

JS insertUsers.js

JS setupReplica.js

JS updateData.js

> OUTLINE

> TIMELINE

scripts > JS insertFollowers.js > ...

1 use("socialmedia")

2

3 db.followers.insertMany([

4 { followId:"F1", followerId:"U1", followingId:"U2", followedAt: new Date("2023-06-01") },

5 { followId:"F2", followerId:"U2", followingId:"U3", followedAt: new Date("2023-06-05") },

6 { followId:"F3", followerId:"U3", followingId:"U4", followedAt: new Date("2023-06-10") },

7 { followId:"F4", followerId:"U4", followingId:"U5", followedAt: new Date("2023-06-15") },

8 { followId:"F5", followerId:"U5", followingId:"U1", followedAt: new Date("2023-06-20") }

9 ])

10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27018

rs0 [direct: secondary] test> use socialmedia

switched to db socialmedia

rs0 [direct: secondary] socialmedia> db.followers.find()

[

{

\_id: ObjectId('6a15bff37c22c20407abc114'),

followId: 'F1',

followerId: 'U1',

followingId: 'U2',

followedAt: ISODate('2023-06-01T00:00:00.000Z')

},

{

\_id: ObjectId('6a15bff37c22c20407abc117'),

followId: 'F4',

followerId: 'U4',

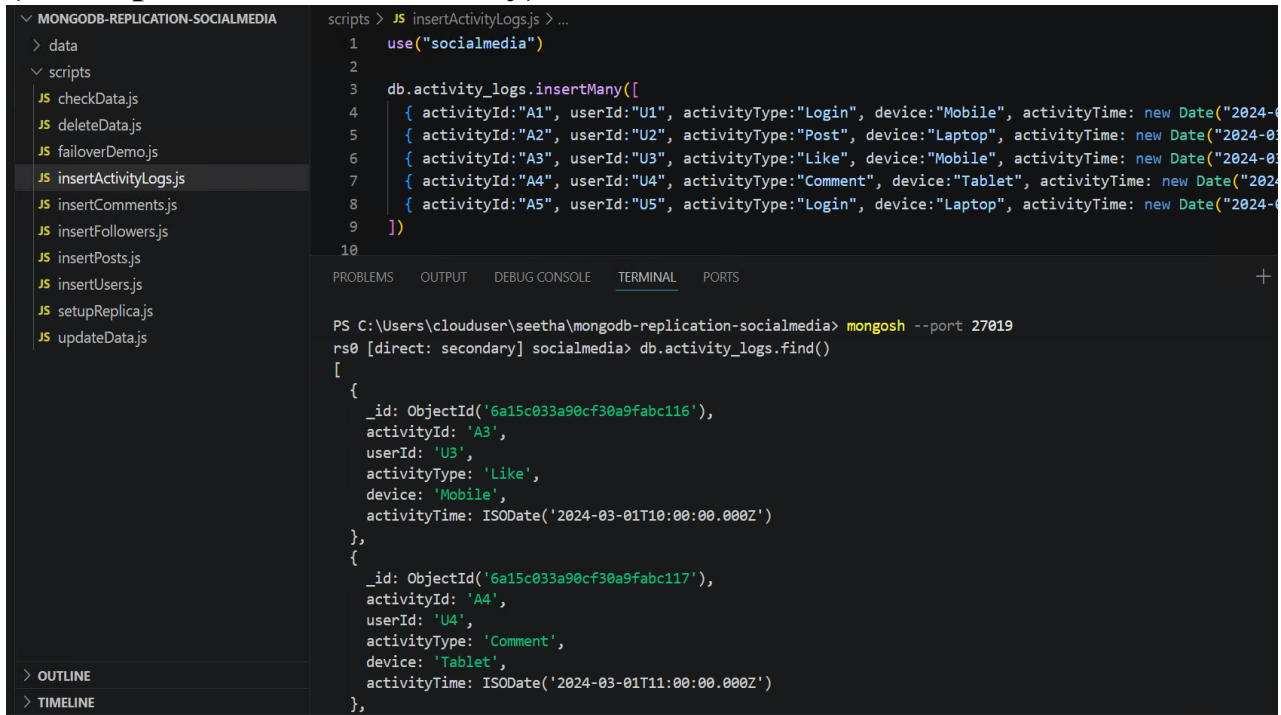
followingId: 'U5',

followedAt: ISODate('2023-06-15T00:00:00.000Z')

},

]

## (From port 27019-secondary)

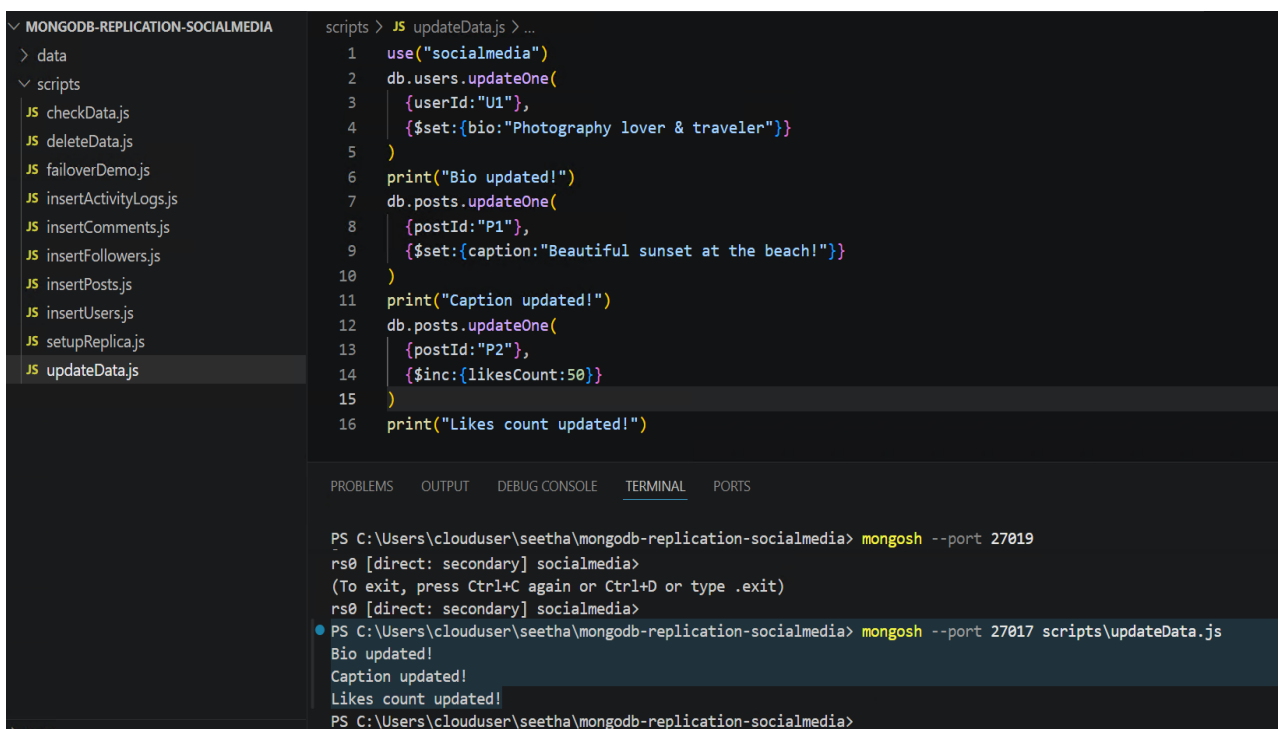


The screenshot shows a VS Code editor with a file explorer on the left containing a project named 'MONGODB-REPLICATION-SOCIALMEDIA'. The 'scripts' folder is expanded, showing several JavaScript files. The 'insertActivityLogs.js' file is selected and its code is visible in the editor. The code uses the MongoDB Node.js driver to connect to a secondary replica set on port 27019 and inserts five activity log documents into the 'activity\_logs' collection. The terminal at the bottom shows the command to run the script using 'mongosh --port 27019', followed by the output of the 'find()' query, which displays the inserted documents in JSON format.

```
scripts > JS insertActivityLogs.js > ...
1 use("socialmedia")
2
3 db.activity_logs.insertMany([
4   { activityId:"A1", userId:"U1", activityType:"Login", device:"Mobile", activityTime: new Date("2024-03-01T10:00:00.000Z") },
5   { activityId:"A2", userId:"U2", activityType:"Post", device:"Laptop", activityTime: new Date("2024-03-01T11:00:00.000Z") },
6   { activityId:"A3", userId:"U3", activityType:"Like", device:"Mobile", activityTime: new Date("2024-03-01T12:00:00.000Z") },
7   { activityId:"A4", userId:"U4", activityType:"Comment", device:"Tablet", activityTime: new Date("2024-03-01T13:00:00.000Z") },
8   { activityId:"A5", userId:"U5", activityType:"Login", device:"Laptop", activityTime: new Date("2024-03-01T14:00:00.000Z") }
9 ])
10
```

```
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27019
rs0 [direct: secondary] socialmedia> db.activity_logs.find()
[
  {
    _id: ObjectId('6a15c033a90cf30a9fab116'),
    activityId: 'A3',
    userId: 'U3',
    activityType: 'Like',
    device: 'Mobile',
    activityTime: ISODate('2024-03-01T12:00:00.000Z')
  },
  {
    _id: ObjectId('6a15c033a90cf30a9fab117'),
    activityId: 'A4',
    userId: 'U4',
    activityType: 'Comment',
    device: 'Tablet',
    activityTime: ISODate('2024-03-01T13:00:00.000Z')
  },
  {
    _id: ObjectId('6a15c033a90cf30a9fab118'),
    activityId: 'A5',
    userId: 'U5',
    activityType: 'Login',
    device: 'Laptop',
    activityTime: ISODate('2024-03-01T14:00:00.000Z')
  },
  {
    _id: ObjectId('6a15c033a90cf30a9fab119'),
    activityId: 'A2',
    userId: 'U2',
    activityType: 'Post',
    device: 'Laptop',
    activityTime: ISODate('2024-03-01T11:00:00.000Z')
  },
  {
    _id: ObjectId('6a15c033a90cf30a9fab11a'),
    activityId: 'A1',
    userId: 'U1',
    activityType: 'Login',
    device: 'Mobile',
    activityTime: ISODate('2024-03-01T10:00:00.000Z')
  }
]
```

## UPDATE



The screenshot shows the same VS Code editor with the 'updateData.js' file selected in the 'scripts' folder. The code uses the MongoDB Node.js driver to connect to a secondary replica set on port 27019 and updates three documents: a user's bio, a post's caption, and a post's like count. The terminal shows the command to run the script using 'mongosh --port 27017 scripts\updateData.js' (note the typo in the original image), followed by the output messages: 'Bio updated!', 'Caption updated!', and 'Likes count updated!'.

```
scripts > JS updateData.js > ...
1 use("socialmedia")
2 db.users.updateOne(
3   {userId:"U1"},
4   {$set:{bio:"Photography lover & traveler"}}
5 )
6 print("Bio updated!")
7 db.posts.updateOne(
8   {postId:"P1"},
9   {$set:{caption:"Beautiful sunset at the beach!"}}
10 )
11 print("Caption updated!")
12 db.posts.updateOne(
13   {postId:"P2"},
14   {$inc:{likesCount:50}}
15 )
16 print("Likes count updated!")
```

```
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27019
rs0 [direct: secondary] socialmedia>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
rs0 [direct: secondary] socialmedia>
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\updateData.js
Bio updated!
Caption updated!
Likes count updated!
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>
```

# DELETE

```
▼ MONGODB-REPLICATION-SOCIALMEDIA
  > data
  ▼ scripts
    JS checkData.js
    JS deleteData.js
    JS failoverDemo.js
    JS insertActivityLogs.js
    JS insertComments.js
    JS insertFollowers.js
    JS insertPosts.js
    JS insertUsers.js
    JS setupReplica.js
    JS updateData.js

scripts > JS deleteData.js > ...
1 use("socialmedia")
2 db.users.deleteOne({ userId: "U5" })
3 print("Inactive user deleted!")
4 db.comments.deleteOne({ commentId: "C5" })
5 print("Unwanted comment deleted!")
6 db.posts.deleteOne({ postId: "P5" })
7 print("Post deleted!")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017
rs0 [direct: primary] socialmedia>
● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\deleteData.js
Inactive user deleted!
Unwanted comment deleted!
Post deleted!
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>
```

## failover.js

```
▼ MONGODB-REPLICATION-SOCIALMEDIA
  > data
  ▼ scripts
    JS checkData.js
    JS deleteData.js
    JS failoverDemo.js
    JS insertActivityLogs.js
    JS insertComments.js
    JS insertFollowers.js
    JS insertPosts.js
    JS insertUsers.js
    JS setupReplica.js
    JS updateData.js

scripts > JS failoverDemo.js > ...
1 use("socialmedia")
2 print("=== Current Replica Set Status ===")
3 rs.status().members.forEach(m => {
4   print(`${m.name} ---> ${m.stateStr}`)
5 })
6 const primary = rs.status().members.find(m => m.stateStr === "PRIMARY")
7 print(`\n=== Current Primary: ${primary.name} ===`)
8 print("Stop this node to trigger automatic election!")
9 print("Then run rs.status() to verify new Primary!")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\failoverDemo.js
=== Current Replica Set Status ===
localhost:27017 ---> PRIMARY
localhost:27018 ---> SECONDARY
localhost:27019 ---> SECONDARY

=== Current Primary: localhost:27017 ===
Stop this node to trigger automatic election!
Then run rs.status() to verify new Primary!
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>

▼ MONGODB-REPLICATION-SOCIALMEDIA
  > data
  ▼ scripts
    JS checkData.js
    JS deleteData.js
    JS failoverDemo.js
    JS insertActivityLogs.js
    JS insertComments.js
    JS insertFollowers.js
    JS insertPosts.js
    JS insertUsers.js
    JS setupReplica.js
    JS updateData.js

scripts > JS failoverDemo.js > ...
1 use("socialmedia")
2 print("=== Current Replica Set Status ===")
3 rs.status().members.forEach(m => {
4   print(`${m.name} ---> ${m.stateStr}`)
5 })
6 const primary = rs.status().members.find(m => m.stateStr === "PRIMARY")
7 print(`\n=== Current Primary: ${primary.name} ===`)
8 print("Stop this node to trigger automatic election!")
9 print("Then run rs.status() to verify new Primary!")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27017 scripts\failoverDemo.js
MongoNetworkError: connect ECONNREFUSED 127.0.0.1:27017
● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27018 scripts\failoverDemo.js
=== Current Replica Set Status ===
localhost:27017 ---> (not reachable/healthy)
localhost:27018 ---> PRIMARY
localhost:27019 ---> SECONDARY

=== Current Primary: localhost:27018 ===
Stop this node to trigger automatic election!
Then run rs.status() to verify new Primary!
PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia>
● PS C:\Users\clouduser\seetha\mongodb-replication-socialmedia> mongosh --port 27019 scripts\failoverDemo.js
=== Current Replica Set Status ===
localhost:27017 ---> PRIMARY
localhost:27018 ---> (not reachable/healthy)
localhost:27019 ---> SECONDARY

=== Current Primary: localhost:27017 ===
```